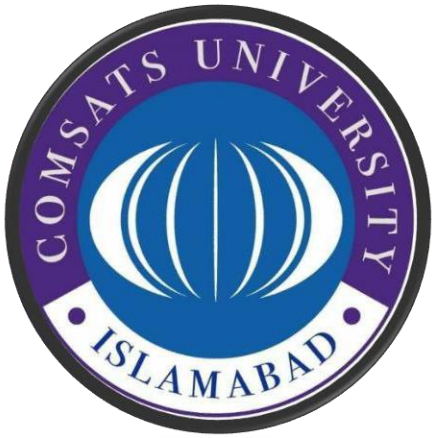
“Web Technologies”

Lab Mid-Term

29th October, 2026



Submitted To:

M. Usman Akram

Submitted By:

Fateh Ali (FA24-BCS-032)

Department **of Computer Science**

COMSATS University Islamabad

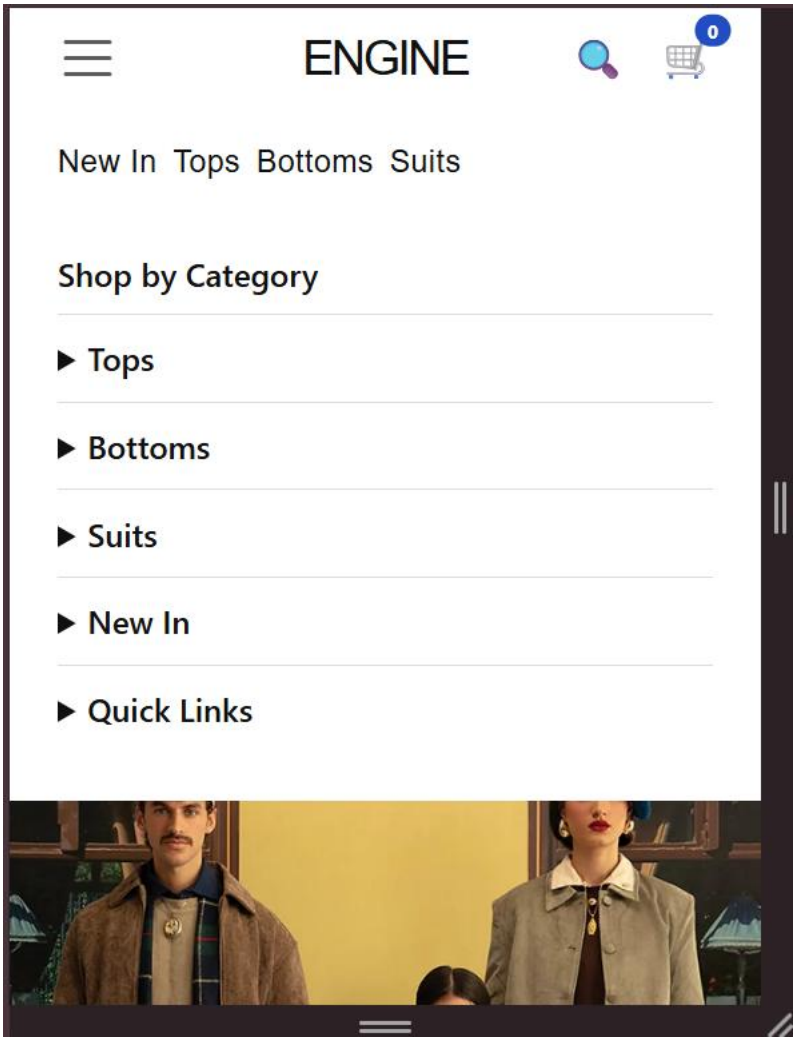
Lahore Campus

LAB MID-TERM

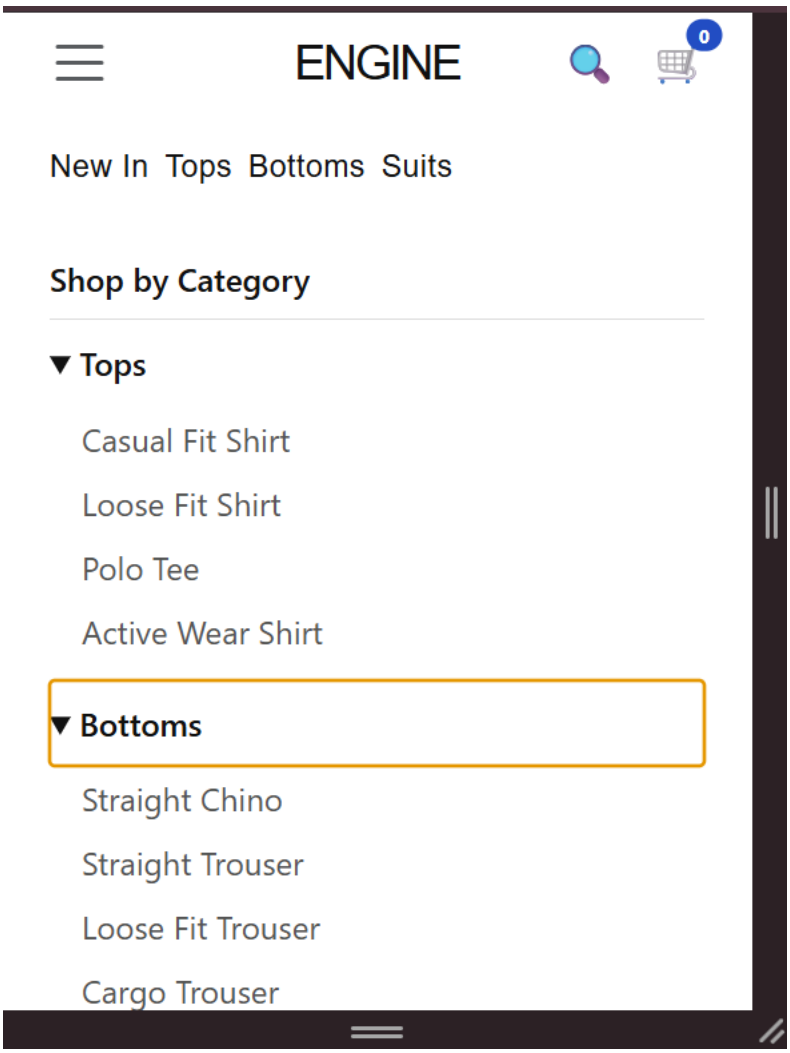
Screenshots:

Task 1: Mega-menu

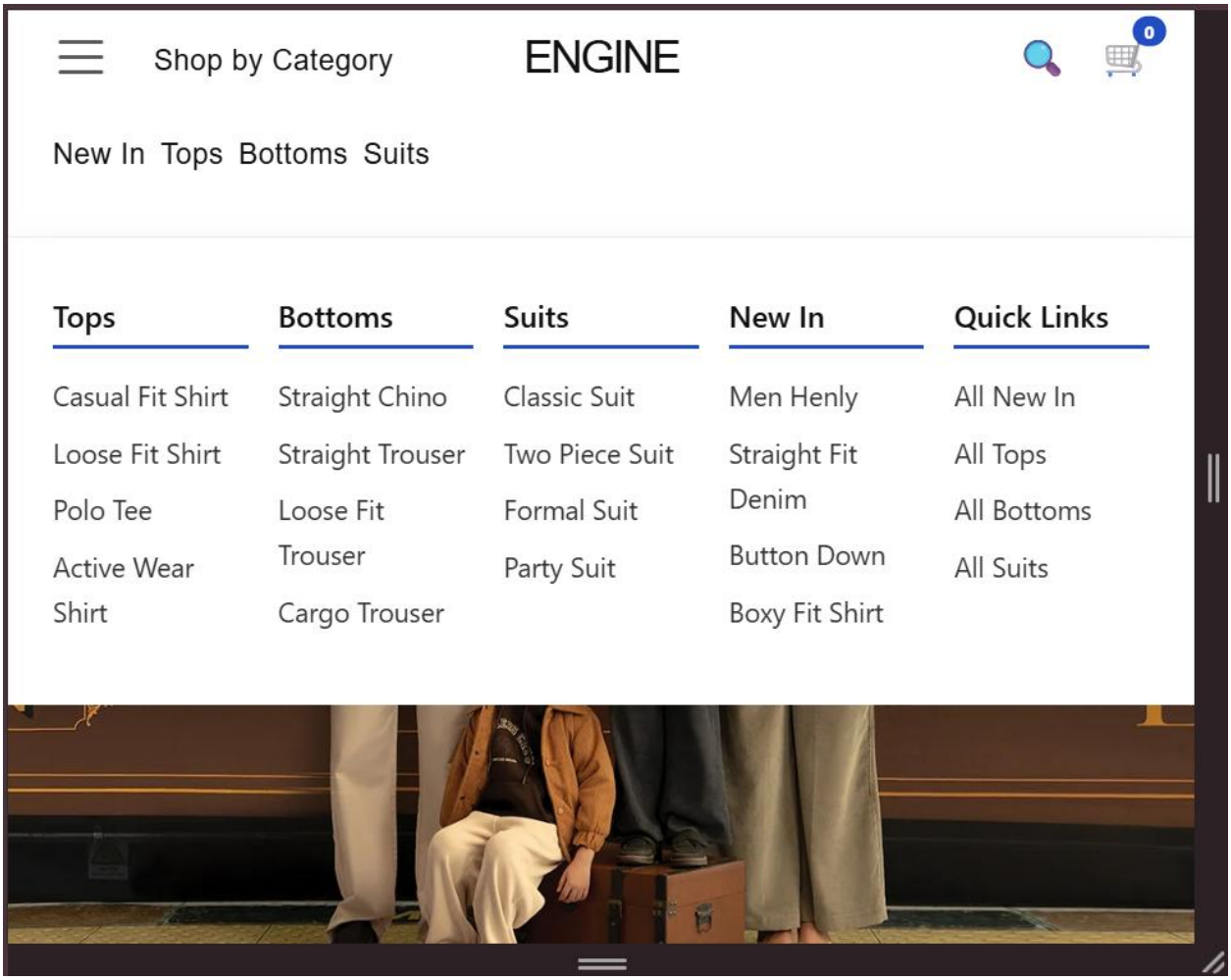
375px: Hamburger menu open, with the “Shop by Category” accordions all closed



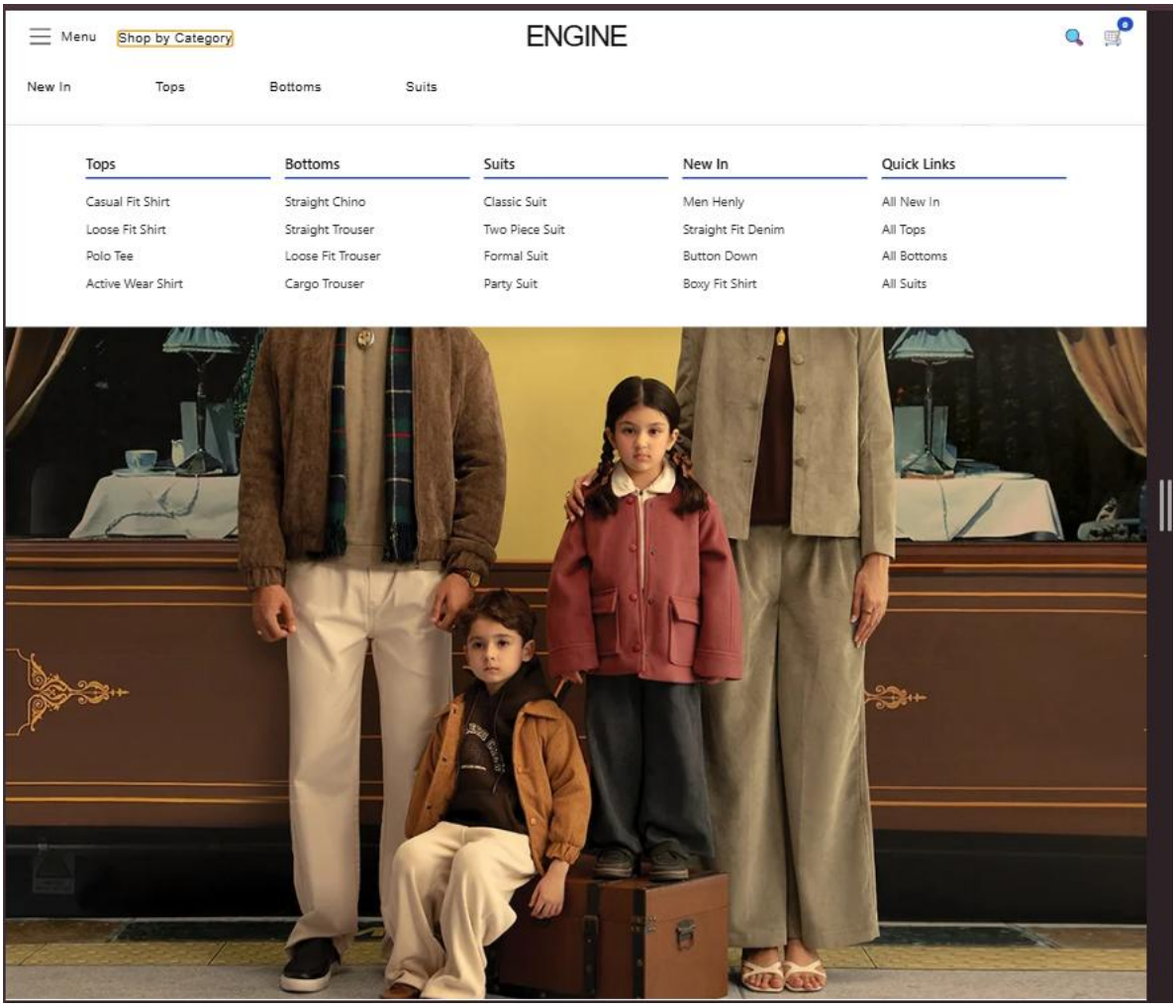
375px: The same menu with two groups open (for example Tops and Suits), to show they open independently



632px: Panel open with all 5 columns, headings and links visible

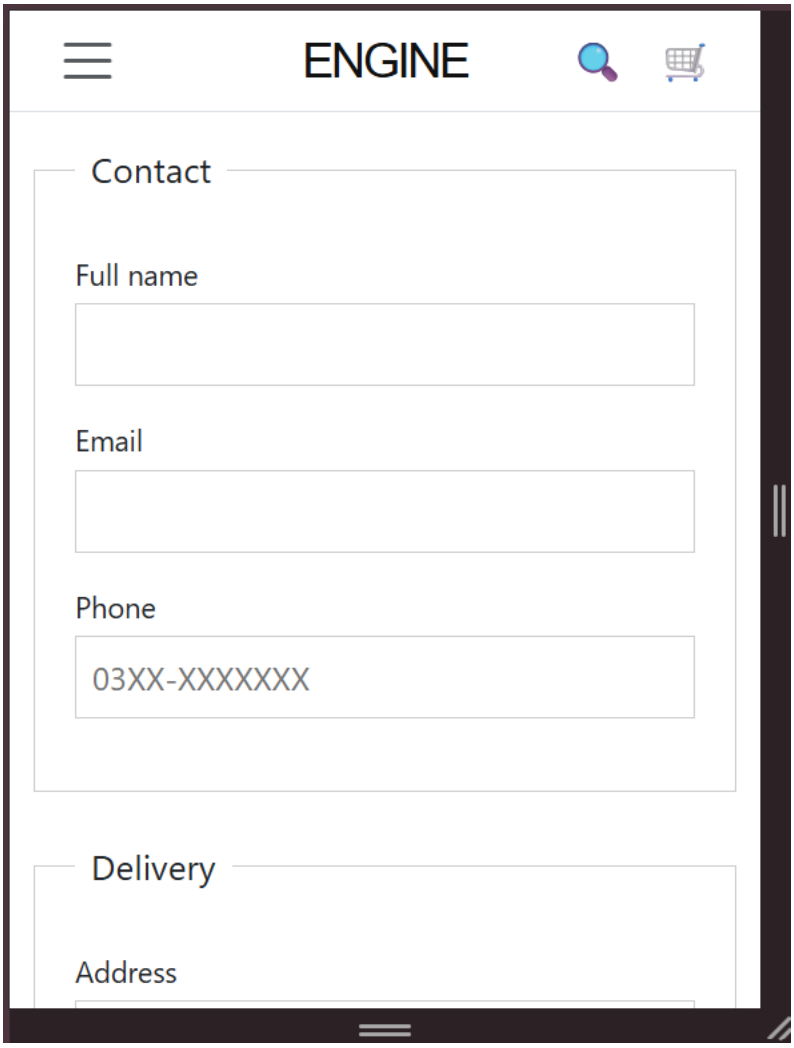


1280px: Panel open over the product grid or the hero, to show it sits above the page

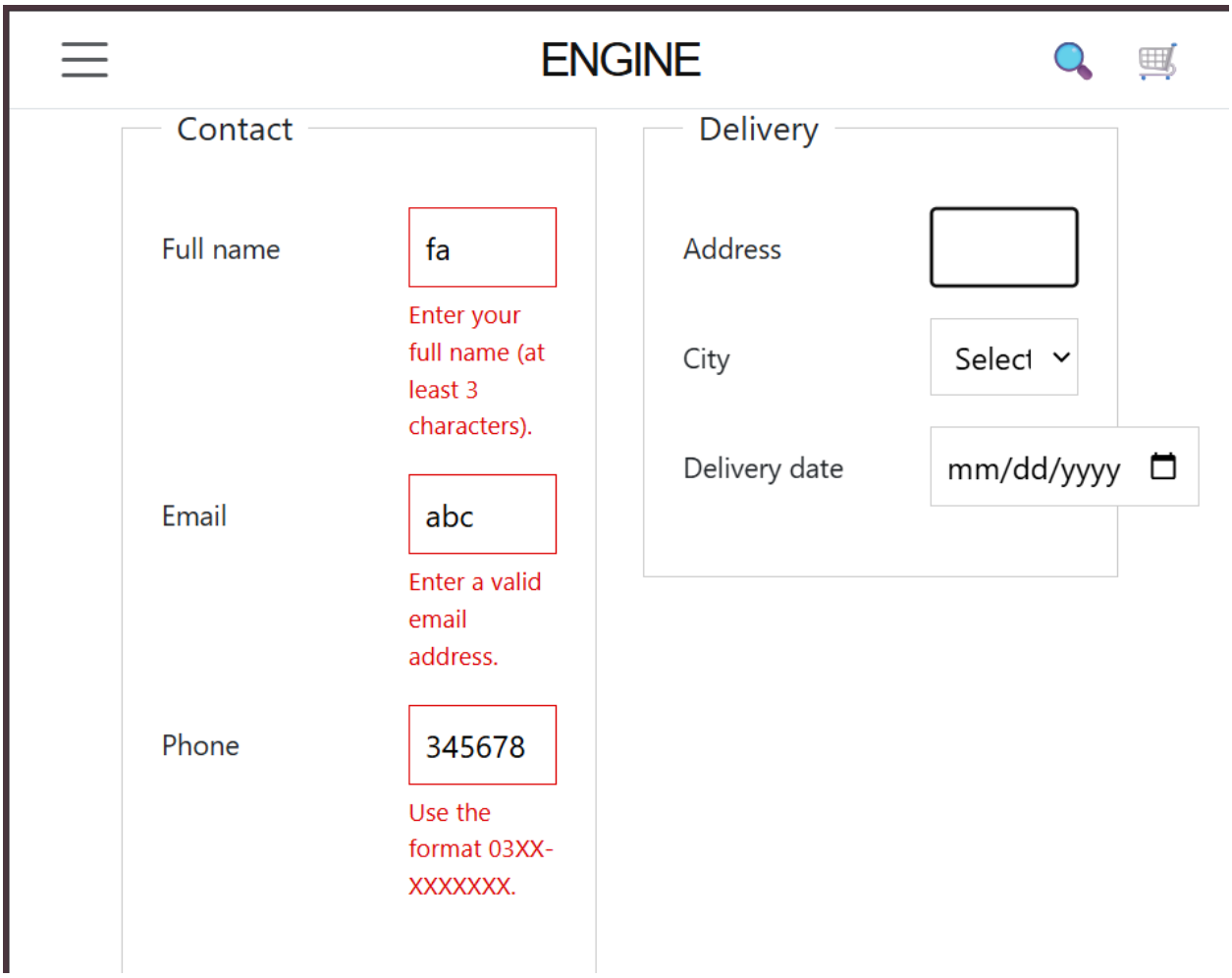


Task 2: Checkout

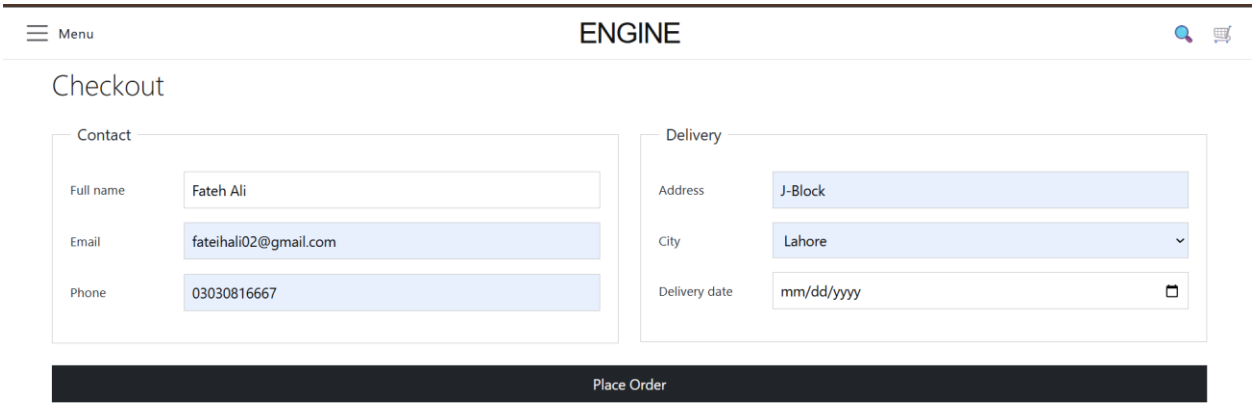
375px: checkout.html just loaded: one column, labels above inputs, no red errors, full-width button



632px: Two fieldsets side by side, labels in a left column, with some errors showing

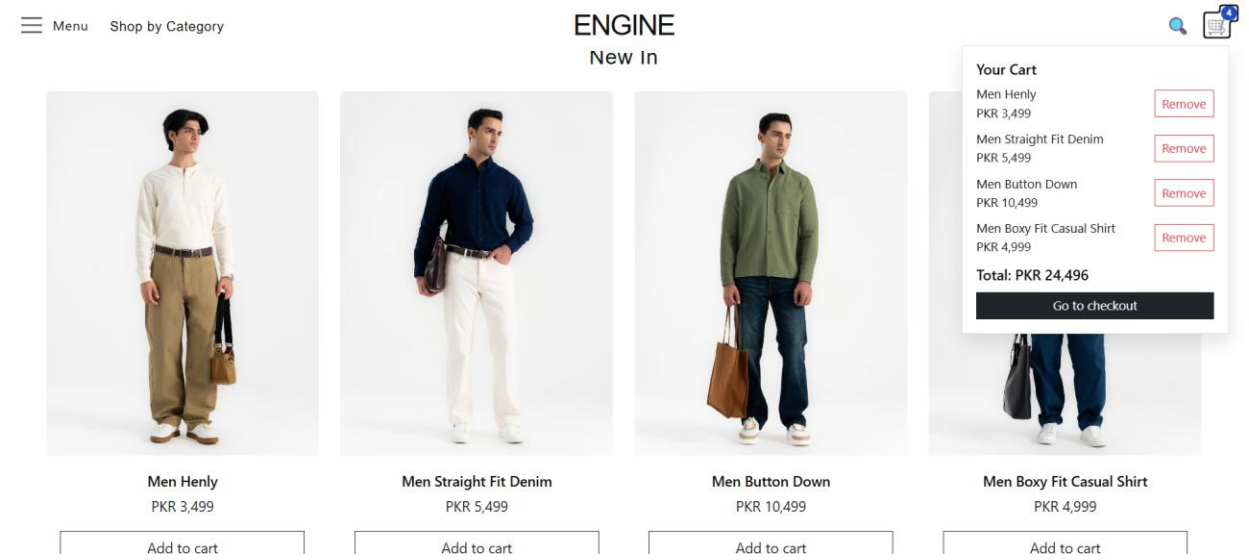


1280px: The same layout with the form filled in correctly and no red

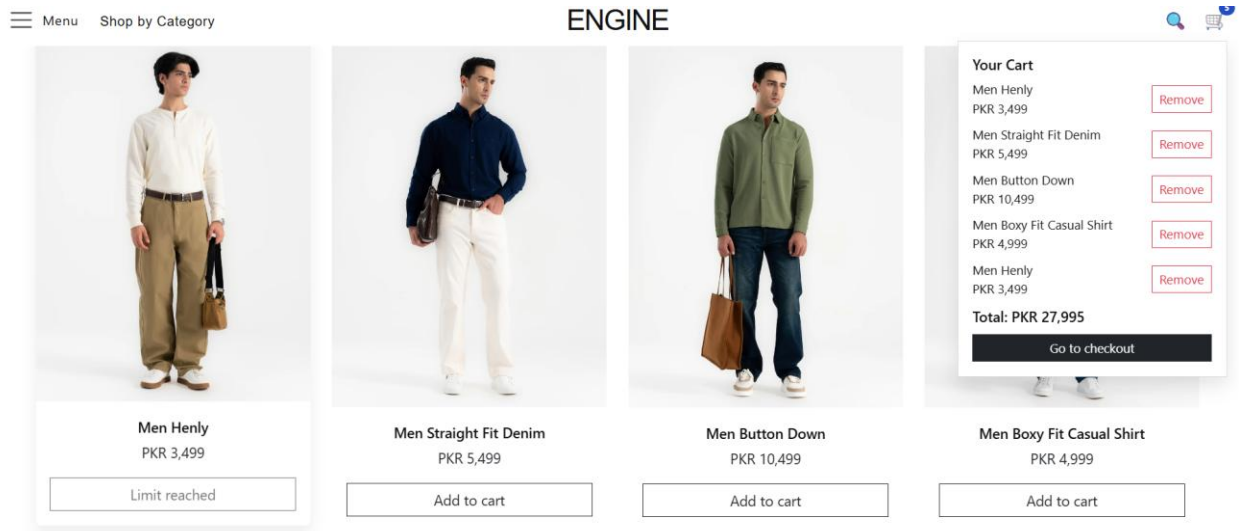


Task 3: Cart

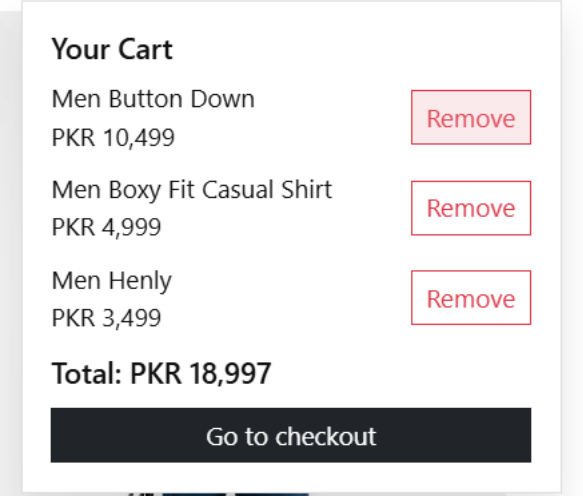
Cart Panel open with items



A card showing “Limit reached”



The cart after clicking Remove on one item, with the badge and total updated



Updated Code:

Task 1:

index.html:

```
<div class="mega-item">
              <button type="button" class="mega-trigger" aria-
haspopup="true">
                Shop by Category
              </button>
              <div class="mega-panel">
```

```
                <div class="mega-grid">
                  <div class="mega-col">
                    <h3 class="mega-heading">Tops</h3>
                    <ul>
                      <li><a href="index.html#tops">Casual Fit
Shirt</a></li>
                      <li><a href="index.html#tops">Loose Fit
Shirt</a></li>
                      <li><a href="index.html#tops">Polo Tee</a></li>
                      <li><a href="index.html#tops">Active Wear
Shirt</a></li>
                    </ul>
                  </div>
                  <div class="mega-col">
                    <h3 class="mega-heading">Bottoms</h3>
                    <ul>
                      <li><a href="index.html#bottoms">Straight
Chino</a></li>
                      <li><a href="index.html#bottoms">Straight
Trouser</a></li>
                      <li>
                        <a href="index.html#bottoms">Loose Fit Trouser</a>
                      </li>
                      <li><a href="index.html#bottoms">Cargo
Trouser</a></li>
                    </ul>
                  </div>
                  <div class="mega-col">
                    <h3 class="mega-heading">Suits</h3>
                    <ul>
                      <li><a href="index.html#suits">Classic Suit</a></li>
                      <li><a href="index.html#suits">Two Piece
Suit</a></li>
                      <li><a href="index.html#suits">Formal Suit</a></li>
                      <li><a href="index.html#suits">Party Suit</a></li>
                    </ul>
                  </div>
                  <div class="mega-col">
                    <h3 class="mega-heading">New In</h3>
                    <ul>
                      <li><a href="index.html#new-in">Men Henly</a></li>
                      <li>
                        <a href="index.html#new-in">Straight Fit Denim</a>
```

```
                    </li>
                    <li><a href="index.html#new-in">Button Down</a></li>
                    <li><a href="index.html#new-in">Boxy Fit
Shirt</a></li>
                  </ul>
                </div>
                <div class="mega-col">
                  <h3 class="mega-heading">Quick Links</h3>
                  <ul>
                    <li><a href="index.html#new-in">All New In</a></li>
                    <li><a href="index.html#tops">All Tops</a></li>
                    <li><a href="index.html#bottoms">All
Bottoms</a></li>
                    <li><a href="index.html#suits">All Suits</a></li>
                  </ul>
                </div>
              </div>
            </div>
<div class="mega-mobile">
            <p class="mega-mobile-title">Shop by Category</p>
            <details>
              <summary>Tops</summary>
              <ul>
                <li><a href="#tops">Casual Fit Shirt</a></li>
                <li><a href="#tops">Loose Fit Shirt</a></li>
                <li><a href="#tops">Polo Tee</a></li>
                <li><a href="#tops">Active Wear Shirt</a></li>
              </ul>
            </details>
            <details>
              <summary>Bottoms</summary>
              <ul>
                <li><a href="#bottoms">Straight Chino</a></li>
                <li><a href="#bottoms">Straight Trouser</a></li>
                <li><a href="#bottoms">Loose Fit Trouser</a></li>
                <li><a href="#bottoms">Cargo Trouser</a></li>
              </ul>
            </details>
            <details>
              <summary>Suits</summary>
              <ul>
                <li><a href="#suits">Classic Suit</a></li>
                <li><a href="#suits">Two Piece Suit</a></li>
```

```
              <li><a href="#suits">Formal Suit</a></li>
              <li><a href="#suits">Party Suit</a></li>
            </ul>
          </details>
          <details>
            <summary>New In</summary>
            <ul>
              <li><a href="#new-in">Men Henly</a></li>
              <li><a href="#new-in">Straight Fit Denim</a></li>
              <li><a href="#new-in">Button Down</a></li>
              <li><a href="#new-in">Boxy Fit Shirt</a></li>
            </ul>
          </details>
          <details>
            <summary>Quick Links</summary>
            <ul>
              <li><a href="#new-in">All New In</a></li>
              <li><a href="#tops">All Tops</a></li>
              <li><a href="#bottoms">All Bottoms</a></li>
              <li><a href="#suits">All Suits</a></li>
            </ul>
          </details>
        </div>
```

Style.css:

```
:root {
  --accent: hsl(224, 70%, 45%);
  --k: 5;
}

/* ===== TASK 1: mega-menu ===== */
.mega-item {
  display: none;
  margin-left: 1.5rem;
}
.mega-mobile {
  width: 100%;
  padding-bottom: 1rem;
}
.mega-mobile-title {
  font-weight: 600;
  margin: 0.5rem 0;
```

```
  color: #111;
}
.mega-mobile details {
  border-top: 1px solid #ddd;
}
.mega-mobile summary {
  padding: 0.6rem 0;
  font-weight: 500;
  color: #111;
  cursor: pointer;
}
.mega-mobile ul {
  list-style: none;
  padding-left: 1rem;
  margin: 0 0 0.5rem;
}
.mega-mobile a {
  display: block;
  padding: 0.25rem 0;
  color: #555;
  text-decoration: none;
}

.custom-navbar .mega-trigger {
  background: none;
  border: 0;
  padding: 0;
  font:
    500 15px Arial,
    Helvetica,
    sans-serif;
  letter-spacing: 0.5px;
  white-space: nowrap;
  cursor: pointer;
  color: #fff !important;
}
.custom-navbar:hover .mega-trigger,
.custom-navbar.scrolled .mega-trigger,
.custom-navbar:has(.show) .mega-trigger {
  color: #111 !important;
}

@media (min-width: 632px) {
```

```css
.mega-item {
  display: block;
}
.mega-mobile {
  display: none;
}

.mega-panel {
  position: absolute;
  left: 0;
  right: 0;
  top: 100%;
  z-index: 1100;
  background: #fff;
  border-top: 1px solid #eee;
  box-shadow: 0 12px 24px rgba(0, 0, 0, 0.1);
  padding: 2rem 1.5rem;
  opacity: 0;
  visibility: hidden;
  transition:
    opacity 0.15s,
    visibility 0.15s;
}
.mega-panel::before {
  content: "";
  position: absolute;
  left: 0;
  right: 0;
  bottom: 100%;
  height: 1.5rem;
}
.mega-item:hover .mega-panel,
.mega-item:focus-within .mega-panel {
  opacity: 1;
  visibility: visible;
}

.mega-grid {
  display: grid;
  grid-template-columns: repeat(var(--k), 1fr);
  gap: 1rem;
  max-width: 1100px;
  margin: 0 auto;
```

```
  }
  .mega-heading {
    font-size: 1rem;
    font-weight: 600;
    color: #111;
    border-bottom: 2px solid var(--accent);
    padding-bottom: 0.4rem;
    margin-bottom: 0.75rem;
  }
  .mega-col ul {
    list-style: none;
    padding: 0;
    margin: 0;
  }
  .mega-col li {
    margin-bottom: 0.4rem;
  }
  .mega-col a {
    color: #333;
    text-decoration: none;
    font-size: 0.9rem;
  }
  .mega-col a:hover {
    text-decoration: underline;
    color: var(--accent);
  }
}
```

Task 2:

Index and product.html

```
<a href="#checkout.html" class="text-decoration-none fs-5 text-dark"
              >🛒</a
            >
```

Checkout.html:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0"
/>
```

```
    <title>Checkout - Engine</title>

    <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstra
p.min.css"
      rel="stylesheet"
      integrity="sha384-
sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB"
      crossorigin="anonymous"
    />
    <link
      rel="stylesheet"
      href="https://cdn.jsdelivr.net/npm/bootstrap-
icons@1.11.3/font/bootstrap-icons.min.css"
    />
    <link rel="stylesheet" href="style.css" />
  </head>
  <body>
    <nav class="navbar sticky-top custom-navbar scrolled border-bottom">
      <div class="container-fluid flex-wrap px-4 align-items-start">
        <div class="d-flex w-100 justify-content-between align-items-
center">
          <div class="nav-section">
            <button
              class="navbar-toggler border-0 p-0 d-flex align-items-center
gap-2 shadow-none"
              type="button"
              data-bs-toggle="collapse"
              data-bs-target="#mainMenu"
            >
              <span class="navbar-toggler-icon"></span>
              <span class="menu-label d-none d-md-inline">Menu</span>
            </button>
          </div>

          <div class="nav-section center-section text-section">
            <a
              class="navbar-brand custom-brand fw-medium m-0 text-
uppercase"
              href="index.html"
              >Engine</a
            >
          </div>
```

```
          <div
            class="nav-section d-flex align-items-center gap-3 justify-
content-end"
          >
            <a href="#" class="text-decoration-none fs-5 text-
dark">🔍</a>
            <a href="#checkout.html" class="text-decoration-none fs-5
text-dark"
              >🛒</a
            >
          </div>
        </div>

        <div class="collapse w-100" id="mainMenu">
          <ul class="navbar-nav fw-semibold gap-2 pb-3 pt-3">
            <li class="nav-item">
              <a class="nav-link" href="index.html#new-in">New In</a>
            </li>
            <li class="nav-item">
              <a class="nav-link" href="index.html#tops">Tops</a>
            </li>
            <li class="nav-item">
              <a class="nav-link" href="index.html#bottoms">Bottoms</a>
            </li>
            <li class="nav-item">
              <a class="nav-link" href="index.html#suits">Suits</a>
            </li>
          </ul>
        </div>
      </div>
    </nav>

    <div class="container py-5 my-4">
      <h2 class="product-title mb-4 pt-3">Checkout</h2>
      <form class="checkout-form" action="#" method="get">
        <div class="checkout-grid">
          <fieldset>
            <legend>Contact</legend>

            <div class="field">
              <label for="fullname">Full name</label>
              <input
```

```
            type="text"
            id="fullname"
            name="fullname"
            required
            minlength="3"
          />
          <small class="msg"
            >Enter your full name (at least 3 characters).</small
          >
        </div>

        <div class="field">
          <label for="email">Email</label>
          <input type="email" id="email" name="email" required />
          <small class="msg">Enter a valid email address.</small>
        </div>

        <div class="field">
          <label for="phone">Phone</label>
          <input
            type="tel"
            id="phone"
            name="phone"
            required
            pattern="03[0-9]{2}-[0-9]{7}"
            placeholder="03XX-XXXXXXX"
            title="Format: 03XX-XXXXXXX"
          />
          <small class="msg">Use the format 03XX-XXXXXXX.</small>
        </div>
      </fieldset>

      <fieldset>
        <legend>Delivery</legend>

        <div class="field">
          <label for="address">Address</label>
          <input
            type="text"
            id="address"
            name="address"
            required
            minlength="5"
```

```
            />
            <small class="msg">Enter your delivery address.</small>
          </div>

          <div class="field">
            <label for="city">City</label>
            <select id="city" name="city" required>
              <option value="" selected disabled>Select city</option>
              <option>Lahore</option>
              <option>Karachi</option>
              <option>Islamabad</option>
              <option>Rawalpindi</option>
            </select>
            <small class="msg">Please choose a city.</small>
          </div>

          <div class="field">
            <label for="ddate">Delivery date</label>
            <input
              type="date"
              id="ddate"
              name="ddate"
              required
              min="2026-10-29"
              max="2026-11-30"
            />
            <small class="msg"
              >Pick a date between 2026-10-29 and 2026-11-30.</small
            >
          </div>
        </fieldset>
      </div>
      <button type="submit" class="btn btn-dark rounded-0 w-100 py-2 mt-
4">
        Place Order
      </button>
    </form>
  </div>
  <footer class="engine-footer py-5 mt-5">
    <div class="container px-4">
      <div class="row g-4">
        <div class="col-12 col-md-6 col-lg-3">
          <h5 class="footer-heading mb-4">Connect With Engine</h5>
```

```
            <div class="footer-contact-item mb-3">
              <i class="bi bi-telephone fs-5 me-3"></i>
              <div>
                <span class="d-block small text-muted">Call</span>
                <strong>042111364463</strong>
              </div>
            </div>

            <div class="footer-contact-item mb-3">
              <i class="bi bi-clock fs-5 me-3"></i>
              <div>
                <span class="d-block small text-muted">Timings</span>
                <strong>Mon - Fri: 09:15AM To 5:30PM</strong>
              </div>
            </div>

            <div class="footer-contact-item mb-3">
              <i class="bi bi-envelope fs-5 me-3"></i>
              <div>
                <span class="d-block small text-muted">Email</span>
                <strong>online@engine.com.pk</strong>
              </div>
            </div>
          </div>

          <div class="col-12 col-md-6 col-lg-3">
            <h5 class="footer-heading mb-4">Support</h5>
            <ul class="list-unstyled footer-links mb-0">
              <li><a href="#">Track Your Order</a></li>
              <li><a href="#">Cart</a></li>
              <li><a href="#">Shipment & Delivery</a></li>
              <li><a href="#">Exchange Policy</a></li>
              <li><a href="#">How to Order</a></li>
              <li><a href="#">Terms & Conditions</a></li>
              <li><a href="#">BaadMay FAQs</a></li>
            </ul>
          </div>

          <div class="col-12 col-md-6 col-lg-3">
            <h5 class="footer-heading mb-4">Company</h5>
            <ul class="list-unstyled footer-links mb-0">
              <li><a href="#">About us</a></li>
```

```
              <li><a href="#">Store Locator</a></li>
              <li><a href="#">Privacy Policy</a></li>
              <li><a href="#">Work with us</a></li>
              <li><a href="#">Payment Method</a></li>
              <li><a href="#">Customer Services</a></li>
              <li><a href="#">Blog News</a></li>
            </ul>
          </div>

          <div class="col-12 col-md-6 col-lg-3">
            <h5 class="footer-heading mb-3">Subscribe to our
newsletter</h5>

            <form class="newsletter-form mb-3">
              <input
                type="email"
                placeholder="Email"
                aria-label="Email"
                required
              />
              <button type="submit"><i class="bi bi-arrow-
right"></i></button>
            </form>

            <p class="small text-muted mb-4">
              Be the first to know about new collections and exclusive
offers.
            </p>

            <div class="social-icons">
              <a href="#" aria-label="Facebook"
                ><i class="bi bi-facebook"></i
              ></a>
              <a href="#" aria-label="Pinterest"
                ><i class="bi bi-pinterest"></i
              ></a>
              <a href="#" aria-label="Instagram"
                ><i class="bi bi-instagram"></i
              ></a>
              <a href="#" aria-label="Snapchat"
                ><i class="bi bi-snapchat"></i
              ></a>
```

```
              <a href="#" aria-label="YouTube"><i class="bi bi-
youtube"></i></a>
              <a href="#" aria-label="LinkedIn"
                ><i class="bi bi-linkedin"></i
              ></a>
            </div>
          </div>
        </div>
        <div class="border-top mt-4 pt-3 small text-center text-muted">
          © 2026, Engine. All rights reserved.
        </div>
      </div>
    </footer>

    <script
      src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.
bundle.min.js"
      integrity="sha384-
FKyoEForCGlyvwx9Hj09JcYn3nv7wiPVlz7YYwJrWVcXK/BmnVDxM+D2scQbITxI"
      crossorigin="anonymous"
    ></script>
  </body>
</html>
```

Style.css:

```
.checkout-grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 1.5rem;
}
.checkout-form fieldset {
  border: 1px solid #ccc;
  padding: 1.25rem;
  margin: 0;
  min-width: 0;
}
.checkout-form legend {
  float: none;
  width: auto;
  font-size: 1.1rem;
  padding: 0 0.5rem;
```

```css
}

.field {
  display: grid;
  grid-template-columns: 1fr;
  gap: 0.25rem;
  margin-bottom: 1rem;
}
.field label {
  font-size: 0.9rem;
}
.field input,
.field select {
  width: 100%;
  border: 1px solid #ccc;
  border-radius: 0;
  padding: 0.5rem;
}

.field .msg {
  display: none;
  color: #d00;
  font-size: 0.8rem;
}
.field input:user-invalid,
.field select:user-invalid {
  border-color: #d00;
}
.field input:user-invalid ~ .msg,
.field select:user-invalid ~ .msg {
  display: block;
}

@media (min-width: 632px) {
  .checkout-grid {
    grid-template-columns: 1fr 1fr;
    align-items: start;
  }
  .field {
    grid-template-columns: 7rem 1fr;
    align-items: center;
    column-gap: 1rem;
  }
```

```
  .field .msg {
    grid-column: 2;
  }
}
```

Task 3:

Index.html:

```
<div class="cart-wrap">
              <a
                href="checkout.html"
                id="cart-btn"
                class="text-decoration-none fs-5 position-relative"
                >🛒
                <span
                  id="cart-badge"
                  class="position-absolute top-0 start-100 translate-
middle badge rounded-pill cart-badge"
                  >0</span
                >
              </a>
              <div id="mini-cart" class="mini-cart" hidden>
                <h6 class="mb-2">Your Cart</h6>
                <ul id="cart-items" class="list-unstyled mb-2"></ul>
                <p class="mb-2 fw-semibold">
                  Total: PKR <span id="cart-total">0</span>
                </p>
                <a
                  href="checkout.html"
                  class="btn btn-dark btn-sm rounded-0 w-100"
                  >Go to checkout</a
                >
              </div>
            </div>
```

Style.css:

```
.cart-wrap {
  position: relative;
}
.cart-badge {
  background: var(--accent);
  font-size: 0.6rem;
```

```
}
.mini-cart {
  position: absolute;
  top: 100%;
  right: 0;
  width: 300px;
  max-width: 90vw;
  margin-top: 0.5rem;
  padding: 1rem;
  background: #fff;
  color: #111;
  border: 1px solid #ddd;
  box-shadow: 0 10px 20px rgba(0, 0, 0, 0.1);
  z-index: 1200;
}
```

Script.js:

```
const LIMIT = 2; // (N mod 4) + 2
let cart = [];

const badge = document.getElementById("cart-badge");
const panel = document.getElementById("mini-cart");
const list = document.getElementById("cart-items");
const totalEl = document.getElementById("cart-total");
const cartBtn = document.getElementById("cart-btn");
const addButtons = document.querySelectorAll(".card .btn");

function cardId(card) {
  return card.querySelector("img").getAttribute("src");
}

function countOf(id) {
  return cart.filter((item) => item.id === id).length;
}

function render() {
  badge.textContent = cart.length;

  list.innerHTML = "";
  cart.forEach((item, index) => {
    const li = document.createElement("li");
```

```
    li.className =
      "d-flex justify-content-between align-items-center mb-2 small";
    li.innerHTML =
      "<span>" +
      item.name +
      "<br>PKR " +
      item.price.toLocaleString() +
      "</span>" +
      '<button type="button" class="btn btn-sm btn-outline-danger rounded-
0 remove-btn" data-index="' +
      index +
      '">Remove</button>';
    list.appendChild(li);
  });

  totalEl.textContent = cart
    .reduce((sum, item) => sum + item.price, 0)
    .toLocaleString();

  addButtons.forEach((btn) => {
    const reached = countOf(cardId(btn.closest(".card"))) >= LIMIT;
    btn.disabled = reached;
    btn.textContent = reached ? "Limit reached" : "Add to cart";
  });
}

addButtons.forEach((btn) => {
  btn.addEventListener("click", () => {
    const card = btn.closest(".card");
    const id = cardId(card);
    if (countOf(id) >= LIMIT) return;
    cart.push({
      id: id,
      name: card.querySelector(".card-title").textContent.trim(),
      price: parseInt(
        card.querySelector(".card-text").textContent.replace(/\D/g, ""),
        10,
      ),
    });
    render();
  });
});
```

```
list.addEventListener("click", (e) => {
  if (e.target.matches(".remove-btn")) {
    cart.splice(Number(e.target.dataset.index), 1);
    render();
  }
});

cartBtn.addEventListener("click", (e) => {
  e.preventDefault(); // stay on the page; toggle the panel instead
  panel.hidden = !panel.hidden;
});

render();
```